4

Sequences and Summations

Chapter Learning Outcomes

CLO

4

- Understand the meaning of a sequence.
- Distinguish between the concepts of sequence and series.
- Distinguish between the arithmetic sequence and geometric sequence.
- Calculate the value of the term (i), the common difference (d) and the general term (an) of an arithmetic sequence from information given about
- the sequence.
- Write the explicit and recursive formulas for arithmetic sequence.
- Calculate the value of the term (i), the common ratio (r) and the general term (an) of geometric sequence from information given about
- the sequence.
- Write the explicit and recursive formulas for geometric sequence.
- Determine limits of sequences.
- Find the sum to n terms of an arithmetic series.
- Find the sum to n terms of a geometric series.

Sequence

A **sequence** is a discrete structure used to represent an ordered list.

$$\{3, 8, 13, 18\}$$

What is the difference between a sequence and a series?

Sequence

A **sequence** is defined as an arrangement of numbers **in a** particular order.

$\{3, 8, 13, 18\}$ List with commas

A **series** is defined as the **sum** of the elements of a **sequence**.

$3 + 8 + 13 + 18$ Indicated sum

Arithmetic Sequence

An **Arithmetic Sequence** is defined as a sequence in which there is a common difference (d) between consecutive terms.

{3, 8, 13, 18}

8 - 3 = 5

13 - 8 = 5

18 - 13 = 5

The common difference (d) is : 5

Arithmetic Sequence

EXAMPLE

Which of the following sequences are **arithmetic**? Identify the common difference (d).

$-3, -1, 1, 3, 5, 7, 9, \ldots$

$15.5, 14, 12.5, 11, 9.5, 8, \ldots$

$84, 80, 74, 66, 56, 44, \ldots$

$-8, 6, -4, 2, 0, \ldots$

$-50, -44, -38, -32, -26, \ldots$

Arithmetic Sequence

The general form of an ARITHMETIC sequence.

First Term: a_1

Second Term: $a_2 = a_1 + d$

Third Term: $a_3 = a_1 + 2d$

Fourth Term: $a_4 = a_1 + 3d$

Fifth Term: $a_5 = a_1 + 4d$

nth Term: $\boxed{a_n = a_1 + (n-1)d}$

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Arithmetic Sequence

Formula for the nth term of an ARITHMETIC sequence

$$\boxed{a_n = a_1 + (n-1)d}$$

$a_n =$ The nth term

$a_1 =$ The 1st term

$n =$ The term number

$d =$ The common difference

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Arithmetic Sequence

EXAMPLE

Given: {79, 75, 71, 67, 63, …}. Find a_{32}

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Arithmetic Sequence

EXAMPLE

Given: {79, 75, 71, 67, 63, …}. What term number is **-169**?

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Arithmetic Sequence

EXAMPLE

In an arithmetic sequence, if you know a_1=3.25, a_3=4.25. Find the common difference (d).

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Arithmetic Sequence

EXAMPLE

Find the next four terms of: –9, -2, 5, …

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Important Formulas for Arithmetic Sequence

Explicit Formula

$$a_n = a_1 + (n-1)d$$

Recursive Formula

$$a_n = a_{n-1} + d$$

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Arithmetic Sequence

EXAMPLE

Write the explicit and recursive formula for the sequence: 2,4,6,8,10,…

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Arithmetic Sequence

EXAMPLE

Write the explicit and recursive formula for the sequence: 2,4,6,8,10,…

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Arithmetic Sequence

EXAMPLE

Write the explicit formula for the sequence:

-50,-44,-38,-32,…

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Arithmetic Sequence

EXAMPLE

Write the recursive formula for the sequence:

-50,-44,-38,-32,…

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Geometric Sequence

A **Geometric Sequence** is defined as a sequence in which there is a common ratio (r) between consecutive terms.

Geometric Sequence

EXAMPLE

Which of the following sequences are **geometric**? Identify the common ratio (r).

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Geometric Sequence

The general form of a GEOMETRIC sequence:

First Term:	a_1
Second Term:	$a_2 = a_1 * r$
Third Term:	$a_3 = a_1 * r^2$
Fourth Term:	$a_4 = a_1 * r^3$
Fifth Term:	$a_5 = a_1 * r^4$

nth Term: $a_n = a_1 r^{n-1}$

Geometric Sequence

The general form of a GEOMETRIC sequence:

$$a_n = a_1 r^{n-1}$$

$a_n =$ The nth term

$a_1 =$ The 1st term

$n =$ The term number

r = The common ratio

Geometric Sequence

EXAMPLE

Given: {4, 12, 36, 108, …}. Find a_{10}

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Geometric Sequence

EXAMPLE

Given: {4, 12, 36, 108, …}. What term number is **8,748?**

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Geometric Sequence

EXAMPLE

In a geometric sequence, if you know a_1=4, a_3=36. Find the common ratio (r).

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Geometric Sequence

EXAMPLE

Find the next four terms of: –4, -8, -16, …

Geometric Sequence

-8/-4 = -16/-8 = 2

The common ratio (r) is 2

Multiply by 2 to get next term

Next four terms……**-32, -64, -128, -256**

Important Formulas for Geometric Sequence

Explicit Formula

$$a_n = a_1 r^{n-1}$$

Recursive Formula

$$a_n = a_{n-1} \times r$$

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Geometric Sequence

EXAMPLE

Write the explicit and recursive formula for the sequence:
5, 15, 45, …

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Geometric Sequence

EXAMPLE

Which of the following is a formula for the sequence:

4, -16, 64, …

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Summations

EXAMPLE

Find the summation of the following series:

$1+2+3+4+5+6+7+8+9+10$

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Summations

$$S=1+2+3+4+5+\ldots+n$$

$$S=\sum_{i=1}^{n} i=\frac{n(n+1)}{2}$$

Upper limit

Lower limit

$$\sum_{i=m}^{n} a_i$$

Index of summation

Summations

EXAMPLE

Find the summation of the following series:

$5+6+7+8+ \ldots + 100$

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Some Summation Rules

$$S = \sum_{i=1}^{n} i = \frac{n(n+1)}{2}$$

$$S = \sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$$

$$S = \sum_{i=1}^{n} i^3 = \left[\frac{n(n+1)}{2}\right]^2$$

Summations

EXAMPLE

Find the summation of the following series:

$1+4+9+16+ \ldots + 100$

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Summations

EXAMPLE

Find the summation of the following series:

$1+8+27+64+ \ldots + 1000$

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Sum of Arithmetic Series

Assume S is an arithmetic series with:

- a_1 : **The first term.**
- a_n : **The nth term.**
- n : **Number of terms.**

Then

$$S_n = \frac{n}{2}(a_1 + a_n)$$

Sum of Arithmetic Series

EXAMPLE

Find the summation of the following arithmetic series:

4+11+18+25+32+39

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Sum of Arithmetic Series

EXAMPLE

Find the summation of the following series:

5+6+7+8+ … + 100

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Sum of Arithmetic Series

EXAMPLE

Find the summation of the following series: $\sum_{j=5}^{8}(3j-4).$

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Sum of Arithmetic Series

EXAMPLE

Find the sum of positive integers from 1 to 60.

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Quiz

Find the sum of the following series:

$$5 + 11 + 17 + \ldots + 95$$

$$38 + 35 + 32 + \ldots + 2$$

$$\sum_{n=1}^{7} (2n + 1)$$

$$\sum_{k=3}^{7} (3k + 4)$$

Quiz

Find S_n for each the following arithmetic series:

$$a_1 = 4,\ a_n = 100,\ n = 25$$

$$a_1 = 132,\ d = -4,\ a_n = 52$$

$$a_1 = 40,\ n = 20,\ d = -3$$

$$d = 5,\ n = 16,\ a_n = 72$$

Summation of Geometric Series

Assume S is a geometric series with:

- a1 : The first term.
- r : The common ratio.
- n : Number of terms.

Then

$$\sum_{i=1}^{n} a_1 r^{i-1} = S_n = \frac{a_1(1-r^n)}{1-r} \quad , \quad r \neq 1$$

Summation of Geometric Series

EXAMPLE

Find the summation of the following geometric series:

10+30+90+270

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Check - Chapter Learning Outcomes

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End of Chapter

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